

HUMAN EVOLUTION



MUCH DEBATE HAS SURROUNDED the evolution of humans. However, most scientists are now agreed that modern humans, *Homo sapiens*, are the sole survivors of a number of human species that evolved from the common ancestor of humans and apes some six million years ago. Climatic changes forced our earliest ancestors out of the tropical forests and into open woodlands and grasslands. The challenge of these new habitats resulted in important changes, such as the ability to walk upright and an increase in brain size.

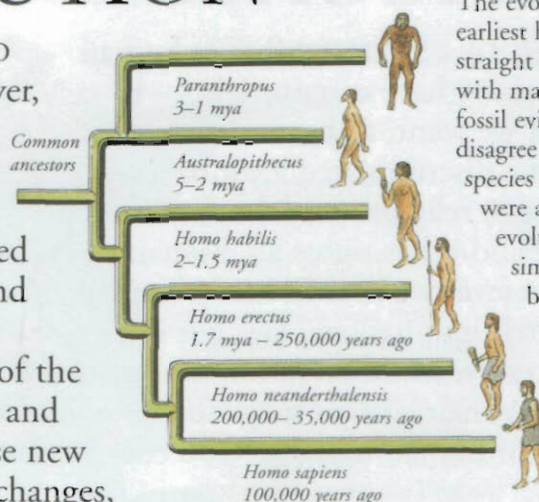


Australopithecines

The Australopithecines are thought to be the earliest hominids (human-like people). Although ape-like, with a small brain and projecting jaws, *Australopithecus* stood upright and walked on two feet. This is known from its leg bones and backbone, and from 3.7 million-year-old footprints found at Laetoli in Tanzania.

Lucy

"Lucy" is the name given to the most complete Australopithecine skeleton yet discovered, found in Ethiopia in 1974. It was an adult female, 3.18 million years old. Lucy was about 1.1 m (3 ft 6 in) tall.



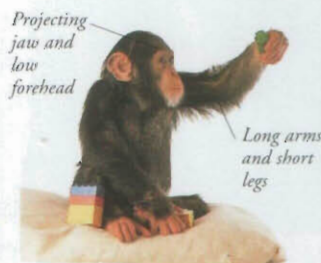
Proconsul

Proconsul is the earliest known member of the hominoids, the group to which apes and humans belong. It lived in the tropical rainforests of East Africa between 24 and 18 million years ago. Compared to its ancestors, *Proconsul* had a large brain.



Proconsul climbed trees and mostly walked on all fours.

Projecting jaw and low forehead



Long arms and short legs

Chimpanzees

The chimpanzee is our closest living relative. Chimpanzees and humans share over 98 per cent of their DNA (genetic material). Chimpanzees and gorillas are known collectively as the African apes. About six million years ago, humans and African apes split from their common ancestor to evolve separately.

Paranthropines had small brains and flattened faces.



Paranthropines

The Paranthropines were strongly built "man-apes" that lived in southern and eastern Africa between three and one million years ago. They were probably descended from the Australopithecines, but were not part of the evolutionary pathway that led to modern humans.

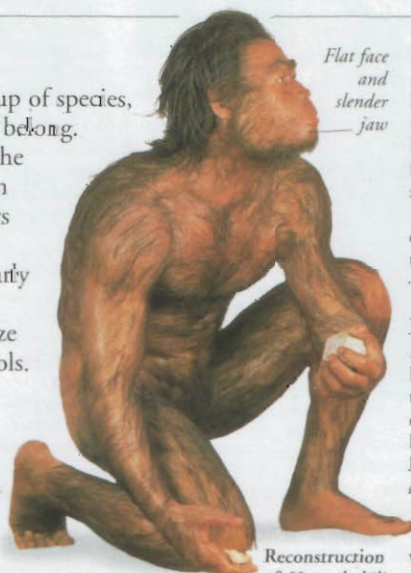
A male *Paranthropus* was about 1.35 m (4 ft 4 in) in height.

Homo

Homo is the genus, or group of species, to which modern humans belong. It probably evolved from the Australopithecines between three and two million years ago, although there is no direct evidence for this. Early members of the genus showed increasing brain size and the ability to make tools.

Homo habilis

"Handy man" is the earliest known species of *Homo*. It lived in the woodlands and savannahs of Africa. *Homo habilis* had a brain size of 650 to 800 ml. It made and used simple stone tools, and was a successful forager and scavenger.



Reconstruction of *Homo habilis*

Flat face and slender jaw

Homo erectus

Homo erectus was the first human to leave Africa and move to Europe and Asia. It had a sloping forehead, flatish face, and a brain size between 850 and 1100 ml. These humans exploited more habitats than their ancestors, and were the first to use fire.



Homo erectus skull

Neanderthals

Neanderthals were the first humans to have adapted to life in the cold climates of Europe and Asia. They had strong physiques and large brains. They wore clothes, made a range of tools, and used fire to keep warm. They were the first humans to bury their dead.



Neanderthal reconstruction

Homo sapiens

Modern humans first evolved in Africa. *Homo sapiens* has a large brain, considerable intelligence, and the ability to use language. Humans increasingly took control of their surroundings as they developed agriculture, societies, and technology.

A modern human has an average brain size of 1400 ml



Face is straight rather than forward-jutting

Prominent brow

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